

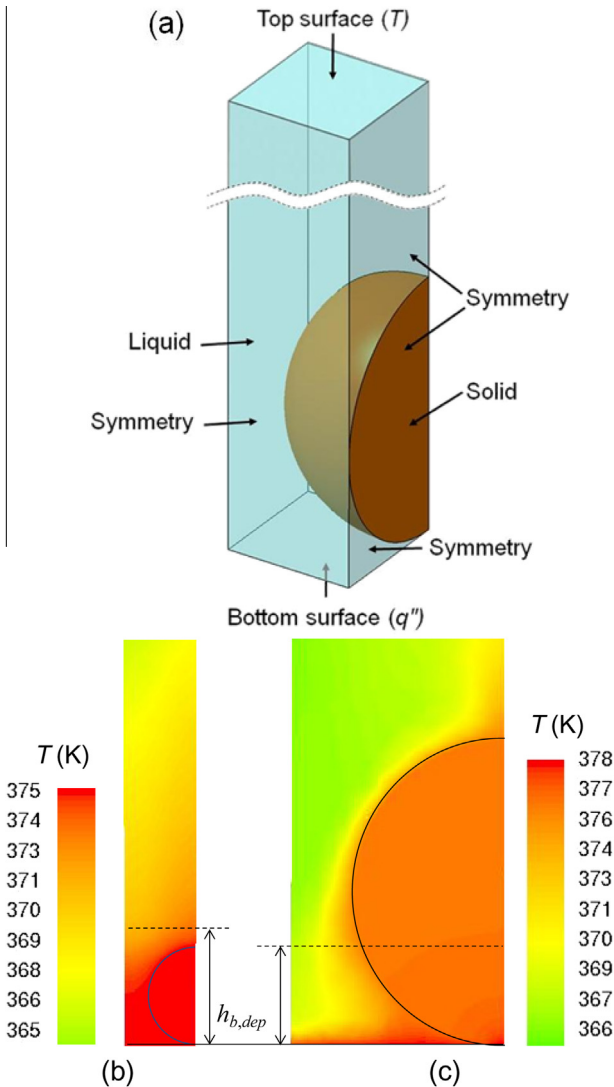


Fig. 6. Numerical simulation: (a) boundary conditions, and results that show the particle and liquid temperature profile, for (b) a 3 mm-diameter free particle at the heat flux of 19.5 kW/m^2 , and for (c) a 9 mm-diameter free particle at 23.5 kW/m^2 . The height of a vapor bubble required for departure from the heated surface, $h_{b,dep}$, obtained from the buoyant versus capillary force balance is superposed.